EMPIRICAL SOFTWARE TESTING · PILOT REPORT

C# LLM-Generated Tests: Coverage vs. Mutation

A deliberately limited, reproducible pilot using one .NET library, local Qwen generation, Coverlet, and Stryker.NET.

> **Scope boundary.** This is a pilot, not a completed eight-library study. Failed generations are reported rather than hidden, and the smoke-test row is excluded from the research comparison.

Subject: Ardalis.GuardClauses · **Target:** GuardAgainstNullExtensions.cs
Date: 8 November 2026 · **Model:** local Qwen · **Temperature:** 0.2

Executive summary

The pipeline successfully restored, compiled, tested, covered, and mutated a valid small suite. However, the ordinary Basic Qwen generations did not compile. Consequently, this pilot cannot answer whether LLM-generated tests show a coverage-versus-mutation gap.

0/4	2.03%	9.09%
Basic suites compiled	Human line coverage	Human mutation score

> **Honest interpretation:** the valid template-based run is an infrastructure smoke test. It supplied the complete test shape and must not be treated as evidence about the Basic LLM strategy.

Research questions

- **RQ1:** Not answerable: Basic Qwen suites failed compilation.
- **RQ2:** Not run: there was no valid Basic suite for feedback.
- **RQ3:** Not run: the pilot does not contain a 60-mutant survivor sample.

Method

The pilot uses the pinned Ardalis.GuardClauses checkout, one target source file, and a separate .NET 8 xUnit project. The human baseline contains three tests. Prompts, raw responses, token counts, and timestamps are stored under `data/raw/llm/`.

Coverlet collects Cobertura line and branch coverage. Stryker.NET 4.8.0 mutates the selected file. Validation records restore status, compiler status, test status, and output tails in `data/results/validation.csv`.

Reproducibility

- Library commit and candidate information are recorded in `libs.csv`.
- Stryker is pinned in `.config/dotnet-tools.json`.
- API configuration is environment-only; no key is stored in code or Git.
- Raw reports and analysis-ready CSV files are retained.

Validation

Run group	Restore	Compile	Pass	Interpretation
Basic Qwen (4)	4/4	0/4	0/4	Not eligible for measurement
Minimal template (1)	1/1	1/1	1/1	Smoke test only

Recorded measurements

Human baseline

Coverlet recorded 12 of 591 lines and 9 of 246 branches covered.

Line coverage · 2.03%

Branch coverage · 3.65%

Stryker created 352 mutants: 5 killed, 3 survived, 47 had no coverage, 2 caused mutant compile errors, and 295 were ignored by filters. The reported mutation score was 9.09%.

Template smoke test

Coverlet recorded 6 of 591 lines and 1 of 246 branches. Stryker reported a 1.82% mutation score. These values verify the measurement path only.

Run	Line	Branch	Mutation
Human baseline	2.03%	3.65%	9.09%
Template smoke test	1.01%	0.40%	1.82%

> **No comparison claim.** The two rows do not represent equivalent test-generation strategies.

Discussion and limitations

The main pilot finding is procedural: compile validity must be established before coverage and mutation scores can be compared. The local Qwen model produced several outputs with Markdown, missing imports, invalid identifiers, incomplete braces, or invalid assertions. Those failures are part of the result.

- One model and one small library limit generalizability.
- Public code may have been present in the model's training data.
- Mutation results depend on Stryker.NET operators and filters.
- No manual survivor classification was performed.
- Coverage values are aggregate assembly values and should not be generalized.

Conclusion

The limited pilot provides no evidence for or against the proposed coverage-versus-mutation relationship. It does provide a reproducible pipeline, visible failure records, and end-to-end measurement smoke-test reports. A larger study should begin only after a genuinely model-generated suite can be produced and validated.

Source artifacts: README.md, data/results/*.csv, data/raw/reports/, and report/report.md.